SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	AI&DS	**Date:**	

Code of Conduct

I _____S.N.V.S Karthik *(*192424186*)__(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ______**karthik________**

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- **Gender and number agreement**
- **Recency**
- **Grammatical role**
- **Semantic compatibility**
- **Discourse coherence**

c) Construct a table showing:

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
He (sentence 2)	Ravi, Arun	Ravi = Valid; Arun = Invalid	Grammatical role (subject preference) and discourse coherence favour Ravi as the continuing topic
him (sentence 3)	Ravi, Arun	Ravi = Valid; Arun = Invalid	Arun is the subject of "helped"; the object pronoun must denote the other participant, Ravi (semantic compatibility)
it (sentence 3)	the book on AI	Valid	Semantic compatibility — "find it" refers to the object being searched for
they (sentence 4)	Ravi and Arun	Valid	Number agreement — plural pronoun requires the combined antecedent set {Ravi, Arun}
the book (sentence 4)	the book on AI	Valid	Repeated definite noun phrase — lexical coreference supported by recency and semantic compatibility

d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

Answer to Q1: Reference Resolution and Discourse Coherence

a) Referring Expressions and Possible Antecedents:

• "He" (sentence 2) — possible antecedents: Ravi, Arun

• "him" (sentence 3) — possible antecedents: Ravi, Arun

• "it" (sentence 3) — possible antecedent: a book on Artificial Intelligence

• "they" (sentence 4) — possible antecedent: Ravi and Arun (combined)

• "the book" (sentence 4) — possible antecedent: a book on Artificial Intelligence

b) Applying the Constraints:

• Gender and number agreement: "He" and "him" are singular masculine and are compatible with both Ravi and Arun, so this constraint alone cannot disambiguate them. "They" is plural and correctly agrees with the combined set {Ravi, Arun}.

• Recency: Arun is the most recently mentioned entity when "He" appears, so recency alone would wrongly favour Arun; this constraint must be overridden by grammatical role and semantic compatibility.

• Grammatical role: Ravi is the subject of the first sentence, and subjects are preferred antecedents for a following pronoun (subject-preference in centering theory), so "He" is resolved to Ravi.

• Semantic compatibility: In "Arun helped him find it", Arun is the subject performing the action of helping, so the object pronoun "him" must denote the other participant, Ravi, and "it" is compatible only with the entity "book".

• Discourse coherence: Reading the paragraph as Ravi searching for a book with Arun's assistance keeps a coherent Elaboration/Occasion relation across sentences, which is only possible if "He"/"him" consistently refer to Ravi.

c) Coreference Table:

d) Final Coreference Chains:

• Chain 1 (Ravi): Ravi → He → him → they (part of)

• Chain 2 (Arun): Arun → Arun (repeated) → they (part of)

• Chain 3 (Book): a book on Artificial Intelligence → it → the book

e) Rewritten Paragraph:

"Ravi met Arun at the library. Ravi was looking for a book on Artificial Intelligence. Arun helped Ravi find the book. Later, Ravi and Arun discussed the book before leaving."

f) Ambiguity if "He was looking for a book" appeared before Arun was introduced:

If this sentence preceded the introduction of Arun, "He" would have no antecedent available in the prior discourse, since only Ravi and Arun are named and Arun would not yet be mentioned. This creates a dangling/cataphoric reference: the reader cannot resolve "He" until a suitable entity appears later, violating the recency constraint (no earlier candidate exists) and weakening discourse coherence, because the reader must hold the pronoun unresolved and retroactively bind it once Ravi or Arun is introduced, increasing processing difficulty and referential ambiguity.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

1. **Maintain entity coherence using assignment, mistakes, and you.**
2. **Use a Cause–Effect or Elaboration discourse relation.**
3. **Include at least two keywords from:**
    - **review**
    - **improve**
    - **confident**
    - **feedback**
4. **Response length must be 2–3 sentences.**
5. **Maintain a positive and polite tone.**
6. **Do not give contradictory advice.**

Tasks:

a) Identify the dialog acts required.

b) Identify the important entities that must be maintained.

c) Generate three possible responses satisfying all constraints.

d) Compare the responses based on:

- **Coherence**
- **Constraint satisfaction**
- **Politeness**
- **Relevance**

e) Select the best response and justify your answer.

f) Explain what happens if entity coherence is not maintained.

Answer to Q2: Dialog Act Recognition and Response Generation

a) Dialog Acts Required:

• Reassure — an expressive/emotional-support act that reduces the student's anxiety about mistakes.

• Advise — a directive act that recommends a concrete next action (reviewing the work) to address the concern.

b) Important Entities to Maintain:

• assignment — the submitted work being discussed

• mistakes — the errors the student is worried about

• you — the student being addressed, kept consistent throughout the response

c) Three Possible Responses:

Response 1: "Don't worry too much about the assignment — a few mistakes are completely normal. Once you get feedback, review it carefully so you can improve next time."

Response 2: "It's understandable to feel that way, but try to stay confident since the assignment is already submitted. Reviewing your notes now will help you improve before the next one."

Response 3: "You don't need to worry so much; most students make small mistakes in assignments. Ask for feedback so you can review and improve your understanding."

d) Comparison of Responses:

Criteria	Response 1	Response 2	Response 3
Coherence	High — smooth link between reassurance and advice	High — slightly less warm transition	Moderate — reads a little generic
Constraint Satisfaction	All constraints met (entities, 2 keywords, length, tone)	All constraints met (entities, 2 keywords, length, tone)	All constraints met, but tone feels less personal
Politeness	Very polite and encouraging	Polite and empathetic	Polite but slightly blunt
Relevance	Directly addresses assignment and mistakes	Directly addresses the concern	Addresses concern in general terms

e) Best Response and Justification:

Response 1 is the best output. It opens with a direct Reassure act ("a few mistakes are completely normal"), maintains entity coherence across assignment, mistakes and you, uses a clear Cause–Effect relation ("Once you get feedback, review it carefully so you can improve"), includes two required keywords (review, improve), stays within 2–3 sentences, and keeps a warm, polite tone without contradicting itself. Response 2 is coherent but slightly less warm, while Response 3 is acceptable but reads a little more generic/impersonal.

f) Effect of Not Maintaining Entity Coherence:

If entity coherence is not maintained, the response may drift to unrelated topics instead of staying focused on the assignment, the mistakes, and the student, breaking discourse cohesion. This makes the reply feel disjointed and generic, weakens the reassurance because the student cannot see their specific concern being addressed, and can even produce advice that is irrelevant or contradictory to the situation, reducing the overall quality and trustworthiness of the response.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- **A financial institution**
- **The side of a river**

Constraints:

1. **Resolve the correct word sense using contextual information.**
2. **Identify important entities and actions.**
3. **Represent the meaning using predicate logic.**
4. **Preserve semantic relationships.**
5. **Maintain coherence in the generated paraphrase.**

Tasks:

a) Identify the possible meanings of bank.

b) Select the correct sense and justify your decision using contextual constraints.

c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)

d) Generate a simple English paraphrase without ambiguity.

e) Explain how Word Sense Disambiguation improves Machine Translation.

Answer to Q3: Word Sense Disambiguation and Meaning Representation

a) Possible Meanings of "bank":

• Sense 1: A financial institution that handles money, deposits, and loans.

• Sense 2: The side/edge of a river (a raised strip of land bordering a body of water).

b) Correct Sense and Justification:

The correct sense is Sense 2 — the side of a river. This is justified by contextual/selectional constraints: the verb "sat on" requires a physical, sittable surface, which a financial institution cannot be in this literal sense; and the co-occurring words "boats", "moving" and "water" form a semantically coherent scene of a riverside setting. These contextual cues override the more frequent "financial institution" sense and select the riverbank meaning through contextual compatibility.

c) Predicate Logic Representation:

$\exists x\ \exists y\ \exists z\ \exists w\ [\ Priya(x) \wedge bank(y) \wedge sit(x) \wedge beside(x, y) \wedge boat(z) \wedge water(w) \wedge move(z) \wedge across(z, w) \wedge watch(x, z)\ ]$

Informally: $sit(Priya) \wedge bank(y) \wedge beside(Priya, y) \wedge watch(Priya, z) \wedge boat(z) \wedge move(z) \wedge water(w) \wedge across(z, w)$

d) Unambiguous English Paraphrase:

"Priya sat beside the river and watched the boats move across the water."

e) How Word Sense Disambiguation Improves Machine Translation:

WSD allows a machine translation system to select the correct target-language word for an ambiguous source word based on its context, instead of defaulting to the most frequent sense. For example, correctly identifying "bank" as a riverbank rather than a financial institution ensures the system chooses the equivalent target word for "riverbank" instead of "financial bank," which is essential in languages that use entirely different words for the two senses. By resolving ambiguity before translation, WSD reduces mistranslation, preserves the intended meaning, and produces more fluent, contextually accurate translations.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1	3	3	3	3	3	15	75
2	3	3	3	3	3	15	
3	3	3	3	3	3	15	
4	3	3	3	3	3	15	
5	3	3	3	3	3	15	

Faculty In-charge	Course Coordinator	Program Director
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Date: ____________________	Date: ____________________	Date: ____________________